PROPOSAL — Pushing the envelope: deeper 3D + interactive schematics

What we can do NOW on the RTX 4050, what a stronger GPU (RTX 5070, 12 GB) unlocks, and the grounding line we never cross.

3D PARTS

NOW — on your RTX 4050 (grounded)

- **Real-time WebGL (Three.js) viewer**
  Replace flat SVG with true lit 3D: PBR materials (metal/rough), soft shadows, smooth orbit, ambient occlusion. Same geometry, far more recognisable.
- **Parametric TEMPLATE library**
  Beyond 4 primitives: real hex bolt (head+chamfer+thread helix), bearing races, gear teeth from FLIS tooth-count, flanges, multi-hole brackets — driven by real characteristics.
- **Section · measure · exploded assemblies**
  Clipping-plane cross-sections, click-measure, and exploded RPSTL assemblies (figure -> its NSNs arranged in 3D). STL export for reference.

MORE POWER — RTX 5070 12 GB

- **Photogrammetry / Gaussian-splatting from REAL photos**
  Mechanic photographs the actual part; build a TRUE rotatable 3D capture (COLMAP + gsplat/NeRF, offline). VRAM-bound — 6 GB is tight; 12 GB trains faster + higher-res. THIS is the upgrade's headline.
- **AI image->mesh (TripoSR / InstantMesh)**
  One photo -> a mesh in seconds. Fast on GPU, but see the grounding line.

SCHEMATICS

NOW — on your RTX 4050 (grounded)

- **Vectorise raster -> SVG line art**
  Trace each schematic to clean vectors: infinite-zoom crispness, per-wire hover/highlight, recolour. The same drawing, made interactive.
- **Full-text search INSIDE diagrams**
  Once OCR completes, find every sheet containing 'fuel pump relay' or a connector ID; jump straight to it.
- **Callout -> part hotspots**
  Hover a component/figure number -> highlight it and open the NSN (ties to the parts index; OCR-box gated).
- **Cross-sheet / off-page links**
  Click an off-page connector to jump to its continuation sheet; link a connector to its pinout table + the 3D part.

MORE POWER — RTX 5070 12 GB

- **Net / circuit EXTRACTION (the big one)**
  Vision models detect components + wires + labels -> a netlist. Click a wire and the WHOLE circuit lights up across sheets; 'where does J12-7 go?'. Vision-heavy -> more cores/VRAM = faster build + inference over all 1,093 schematics.
- **Symbol recognition + local 'explain this circuit' VLM**
  Identify standard symbols; a small offline vision-language model explains a circuit — grounded to the EXTRACTED netlist, not free invention. 12 GB opens usable VLMs (6 GB can't).

THE GROUNDING LINE (never crossed — it's the whole point of this tool)

GROUNDED (build freely): WebGL rendering, parametric templates, photogrammetry/splats from REAL photos, multi-view reconstruction from REAL drawings, schematic vectorisation + net extraction, OCR search/hotspots — all from real data. FLAGGED (aids only, clearly labelled, never authoritative): single-image AI 3D and AI super-resolution INVENT the hidden sides / unseen strokes. We can offer them as a quick visual aid with a bold 'AI-inferred, not verified' badge — but the 104th sheet and any decision stays on cited, real geometry.

HARDWARE — would the RTX 5070 help?

RTX 4050 (you have)
2,560 CUDA · 6 GB GDDR6
great: OCR, WebGL 3D, vectorise, templates, OCR-search, hotspots

→

RTX 5070 12 GB (Blackwell)
4,608 CUDA · 12 GB GDDR7 · ~798 AI TOPS
unlocks: photogrammetry/Gaussian-splat, net-extraction vision models, local VLM, faster batch over all schematics

Verdict: the 4050 covers the whole 'NOW' column (most of the value). The 5070's 12 GB is worth it specifically for photo->true-3D capture and schematic net-extraction at scale — VRAM is the real limiter, not raw speed.

MY RECOMMENDED ORDER (high value, all grounded, all on the 4050 first)

1. WebGL/Three.js real-3D viewer — biggest visual leap, runs now, makes parts instantly recognisable.
2. Schematic vectorisation + per-wire hover — turns flat scans into interactive line art.
3. Parametric template library — richer, more accurate shapes from the FLIS fields we already hold.
4. (as OCR finishes) full-text schematic search + callout->part hotspots.
5. THEN, if you get the 5070: photo->true-3D capture and schematic net/circuit extraction.

PROPOSAL — your call on order + whether to plan for the 5070. Dark (R3). Nothing built yet.